Creating a Price Prediction and Deal Recommendation Tool for the Austin Metroplex Housing Market using Multivariate Regression Analysis

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https://github.gatech.edu/MGT-6203-Fall-2023-Canvas/Team-5

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Background

The US housing market accounts for approximately 15% of total gross domestic product (GDP). This sizable portion can shift global, national and local economies. Since 2000, there have been four defining periods for the housing market: pre-housing bubble, the burst of the housing bubble (and consequently, The Great Recession), post-recession to early 2020 (defined by the global pandemic), and post pandemic to present day. From 2000-2007, median sales of home prices in the US increased 56% (from $165,300 to $257,400). Following the housing bubble burst and the ensuing recession, median house sale price decreased nearly 20% (from $257,400 to $208,400). The period following the recession was a prosperous one for the housing market. Median sale price increased 49% ($221,100 to $329,000). Finally, following the global pandemic to present day, the housing market has seen explosive growth, 49% in just two years ($322,600 to $479,500). Today, the average sale price of a home in the US sits at about $431,000 (U.S. Census Bureau and U.S. Department of Housing and Urban Development, 2023).

Local housing markets can add an additional layer of complexity. They can be at the mercy of macroeconomic factors and microeconomic factors (local supply, demand, property tax rates, etc.) that also may impact the price of a home. To see all these factors at play, we will examine the housing market in Austin, Texas.

Problem

Austin, TX is the second fastest growing metroplex in the United States (KVUE, 2023) and is among the country's 10 largest cities (City of Austin, 2023). This influx of residents has shifted the supply and demand dynamic in the Austin housing market (Home Buying Institute, 2023). With the current state of the national housing market and the local housing market, prospective home buyers in Austin may have many questions. Is now a good time to buy? Is it better to wait? Is this home a good deal? What about interest rates? What are the features of a home that most contribute to the pricing of said home? Conversely, a prospective seller has a separate set of concerns: is now a good time to sell? Is it better to wait? What are the main drivers of home price? What are the renovations that can be made to increase the value of a home? These dilemmas lead us to our primary objective.

Objective

Our primary objective is to create a tool that local buyers and sellers in Austin can use to confidently assess the value of a home via price prediction. More specifically, our tool will:

1. Instruct on what features **within** the home most contribute to home price
2. Instruct on what features **in the community surrounding the home** most contribute to home price
3. Highlight which communities (if any) in the Austin area are considered more or less desirable
4. Using the difference in model predicted price and actual price, segment properties into one of five recommendations: Strong Buy, Buy, Fairly Priced, Sell or Strong Sell
5. Recommend which home features are worth investing in when renovating the property to maximize the market price of the home, for buyers looking to bolster their asset upon purchase

Hypothesis

We have four hypotheses centered around the specific objectives of our modeling approach and real estate tool:

1. Features **within** the home that will impact the home price will include (but not limited to) square footage, lot size, number of bedrooms, and number of bathrooms
2. Features **in the community surrounding the home** that will impact the home price will include quality of the schools in proximity
3. Using Austin area zip codes as features in the models will highlight both more and less prominent neighborhoods
4. We hypothesize that our prediction-driven segmentation returns an even spread for the five recommendation levels for home listings in the Austin area

Data Collection

Our dataset is a comprehensive dataset from Kaggle. The original dataset consists of over 15,000 data points (home listings) and 45 features from homes in Austin, Texas, from 2018 to 2021. The 45 features consisted of both continuous and categorical variables. Aside from last sale dates and location information, some examples of these fields are:

- hasAssociation - Whether or not the home has homeowner's association
- hasCooling - Whether or not the home has air conditioning
- hasGarage - Whether or not the home has a garage
- hasView - Whether or not the home has a view
- homeType - The type of home that it is
- parkingSpaces - How many parking spaces the home has
- lotSizeSqFt - How many square feet the lot size is

We chose this dataset because of its variety of features and size. On our first iteration of this project, we used a different dataset from Zillow. With that dataset, we found a very strong correlation to the number of bedrooms, but there weren't many other notable features according to the model to predict house price. Therefore, much of the models' meaningful signal wasn't derived from descriptive attributes of the home. This new dataset has a plethora of potentially notable features, and we were able to find much more material results with the variety of factors to play with. The change also gives us the ability to derive significant insights about what physical features make a home's price rise and allows us to recommend renovations or investment opportunities. The link to this dataset can be found in the references at the end of the report.

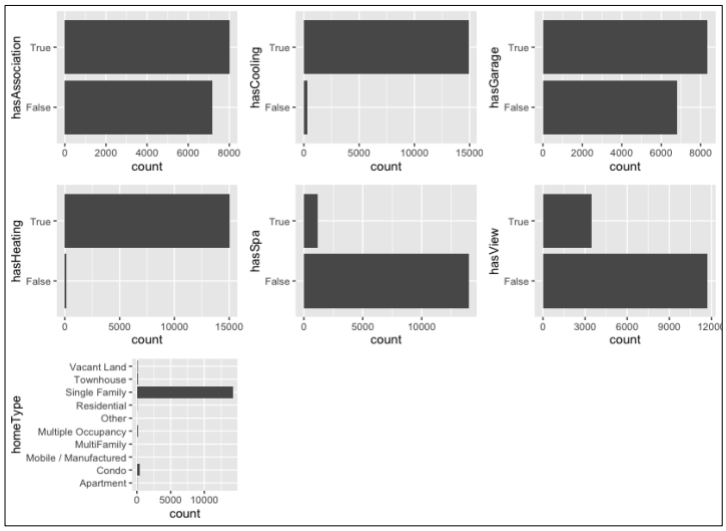
Exploratory Data Analysis

Exploratory Data Analysis (EDA) was the first step in familiarizing ourselves with the dataset. Before diving into some summary statistics, it was important that we check our data for missing/empty values. Fortunately, our data was free from missing data, and we could proceed with the EDA. We looked at the dataset in its entirety by implementing summary statistics, reproduced in a table below.

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
livingAreaSqFt	15171	2208	1379	300	1483	2687	109292
numOfBedrooms	15171	3.4	0.85	0	3	4	20
numOfBathrooms	15171	2.7	1.1	0	2	3	27
numOfStories	15171	1.5	0.52	1	1	2	4
latestPrice	15171	512768	453185	5500	309000	575000	13500000

The summary statistics were particularly useful in becoming familiar with our response variable as well as other continuous variables in the dataset. From this, we have a snapshot of what the average home listing in the Austin market looks like - a 2200 square foot, 3-bedroom, 3-bathroom, 2 story home being listed for $512,000. The caveat comes when looking at the second column of standard deviation. The magnitude of standard deviation for lot size and latest price indicates that while we may have an idea of what the **average** home may look like in Austin, the variety of listings is quite robust. This inference can be verified by looking at the corresponding maximum and minimum values in the table.

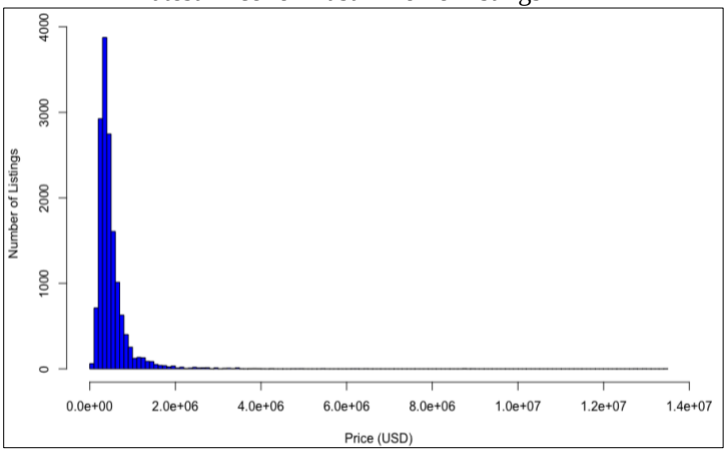
Utilizing a plot grid provided insight into the categorical variables for the listings. The histogram plot grid is pictured below.



Some of the feature distributions represented in the dataset are unsurprising (i.e. most of the listings having heating and cooling). The data set also contains a balanced sample of homes with HOA and garage. One potentially noteworthy finding is that the dataset is overwhelmingly single-family homes. Because of this distribution of home type, the model/tool may be better equipped for feature recommendations and buy/sell suggestions in single family homes than, say, townhouses or apartments.

The last step was a cursory exploration of the response variable, the latest price of the home listings. The price, plotted against the number of listings, is pictured below.

Latest Price for Austin Home Listings



This histogram of price of listings in the Austin dataset is skewed right. The long right tail indicates that there are homes listed for prices that are much higher than the median house price and the mean price is higher than the median as a result. Further testing on the residuals during the modeling phase will provide guidance on if transformation of the inputs, the output, or both, is necessary.

Data Cleaning

Our dataset was not yet in a workable state and needed to be cleaned. To produce a usable data set, we first had to remove any extraneous features. Omitting unnecessary features from the dataset can enhance computational memory and accuracy as well as prevent overfitting on the modeling side. Having fewer features

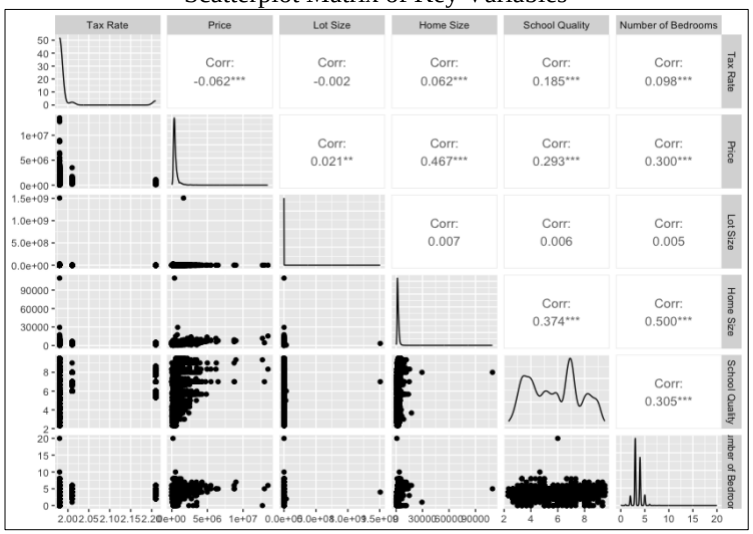
without compromising model accuracy can also make the model easier to understand, explain and implement. As stated above, the uncleaned dataset had 45 predictors (and two additional features with a Zillow id and Zillow image of the property). Not all of these predictors were applicable for our purposes so we thought it best to remove any irrelevant variables. The following features were removed from the dataset: *zpid, description, latest_saledate, latest_salemonth, latest_saleyear, latestPriceSource, homeImage, numPriceChanges, numOfPhotos, city, streetAddress, latitude, longitude*.

We anticipated strong correlation between some of the location related variables (city, streetAddress, zipcode, latitude, longitude) so we retained zipcode and removed the others. A Pearson's chi-square test between zipcode and city returned a p-value of 2.2e-16, allowing us to reject the null hypothesis and confirm our suspicions.

Next, we converted the True/False categorical variables into variables with binary values (1 = True, 0 = False for hasAssociation, hasCooling, hasGarage, hasHeating, hasSpa, and hasView). Zip code and home type were one hot encoded, with each factor level getting its own column and each home listing receiving 1 or 0. The listing would receive a 1 in the column for the zip code where it is located and a 0 for all of the other zip code columns. The listing would also receive a 1 in the column for the corresponding home type and a 0 for all other home type columns. The issue with categorical variables is that some machine learning models do not support them. This one hot encoding technique creates flexibility in modeling the data with more options at our disposal.

The last categorical variable was yearBuilt. Our first instinct was to one hot encode yearBuilt, just like we did with the other categorical variables. After some consideration, we decided that any signal in the year the listing was built might better be captured by a range of years, rather than year to year. For example, the difference in 2012 to 2013 might not be as pronounced as 2001-2010 vs. 2011-2020. For this reason, we created variables for each decade across the dataset from the earliest to the latest. This also served the dataset by utilizing fewer input variables than would otherwise be needed by one hot encoding each year. With the data now fully cleaned, a scatterplot matrix provided insights to some of the key variables pictured below.

Scatterplot Matrix of Key Variables



The scatterplot matrix revealed positive correlation in home size, school quality and number of bedrooms as it relates to price. This lends some support to our early hypotheses about house features as we proceed into our models. Predictors lot size and tax rate do not appear to be highly correlated at this stage in the analysis.

Methodology

We will utilize three modeling techniques: linear regression, random forest regression, and gradient boosted regression. The first model built was the simplest, a **standard multivariable linear regression**. We started simple so that we could begin to interpret some of the basic relationships surfacing between input variables and the target, allowing us to benchmark the performance of more advanced models that we go on to build. Standard multivariable linear regression provides us with easily digestible metrics for both relationships

between input variables and the dependent variable (namely, p-values) and a benchmark for comparison (R-squared for quality of fit and the proportion of variance that can be explained by the input variables). The data was randomly split into train (70% of data points) and test (30% of data points) groups to verify model performance and ensure the models were not overfit.

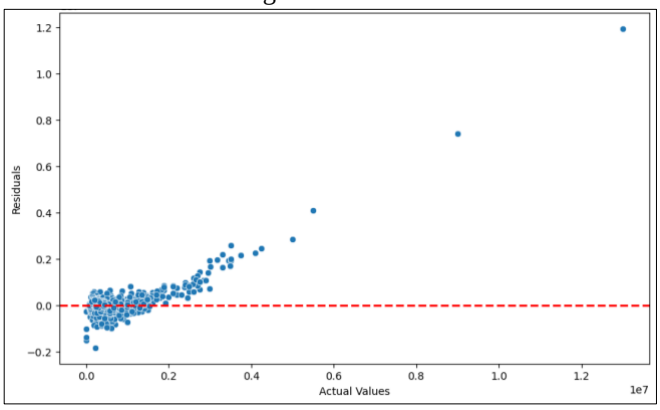
The second model we built was a **Random Forest regression** model. Knowing the merits of a Random Forest approach, such as the ability to efficiently handle large data and deliver better overall estimates, we thought this might yield a more performant model on our data. The data for the Random Forest was randomly split much like it was for the multivariate linear regression model - train (70% of data points) and test (30% of data points) groups. This approach was utilized to help verify our results from the training set using the test set. One of the drawbacks of the random forest approach is the “black box” nature of the model. That is, aside from the result, it is difficult to ascertain what is going on under the hood of a Random Forest. A Random Forest regression works by creating magnitudes of decision trees, all with slightly different data and slightly different sets of parameters. It uses the average response of all the trees in the forest to predict a response. This magnitude makes the Random Forest harder to interpret. There is not a specific model we can utilize to predict a response for a data point and there is no specific model in which we can lean on. The shortcomings of the Random Forest led us to our third model.

The third model we tried was **XGBoost regression**, a machine learning approach that uses an ensemble of decision trees and gradient boosting to make predictions, benefiting from the combined predictive power of multiple learners into a single aggregated model. XGBoost tends to yield higher accuracy, can efficiently handle high row count datasets and other computationally demanding datasets, incorporates regularization techniques to help us avoid overfitting, and makes it easy to grab feature importance scores to tell us how useful each feature was in the construction of the boosted decision trees. Because of its flexibility, XGBoost gives us a lot of choice when tuning hyperparameters, so after splitting our train and test sets, we decided to use Grid search to find the highest performing number of trees and max depth. We also applied 3-fold cross validation on the model to get an even better sense if our model would be performant on unseen data. Instead of linear regression’s coefficients, XGBoost provides some interpretability of the power of certain features by calculating importance for each decision tree by the amount that each attribute split point improves the performance measure, weighted by the number of observations the node is responsible for, and then averaged across all of the decision trees.

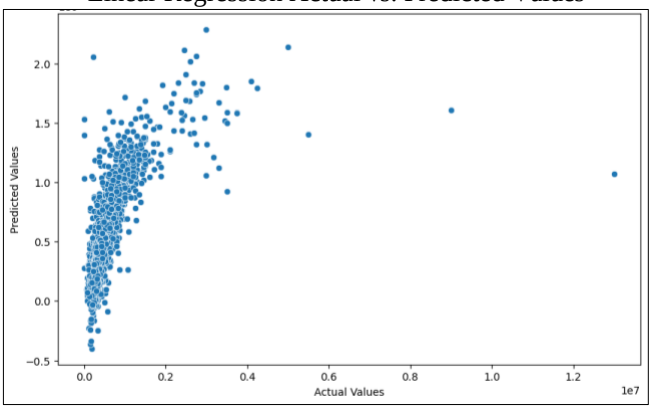
Modeling Results

The **multivariate linear regression**, being the least complex model of the three, performed exactly as expected. This model returned an R^2 value of 0.47, meaning that the model was only able to explain slightly less than half of the variability in Austin housing prices. The root mean squared error for this model, which measures average magnitude between the actual and predicted values, is $353,607. Given the ranging house prices in the dataset, the RMSE for this model suggests that the model performed well in regard to accuracy. Preliminarily, we noticed that the model was slightly struggling to capture the entire dataset, so we log transformed the inputs to help with normality and homoscedasticity. Although this transformation helped the model create more reliable predictions, we will still need to explore more complex models that may be better suited for this dataset.

Linear Regression Residuals Plot

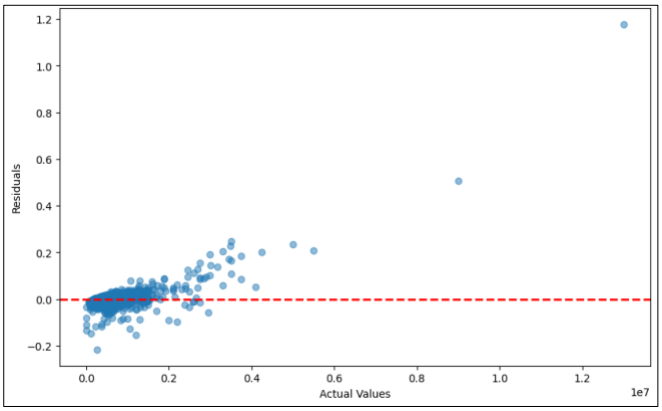


Linear Regression Actual vs. Predicted Values

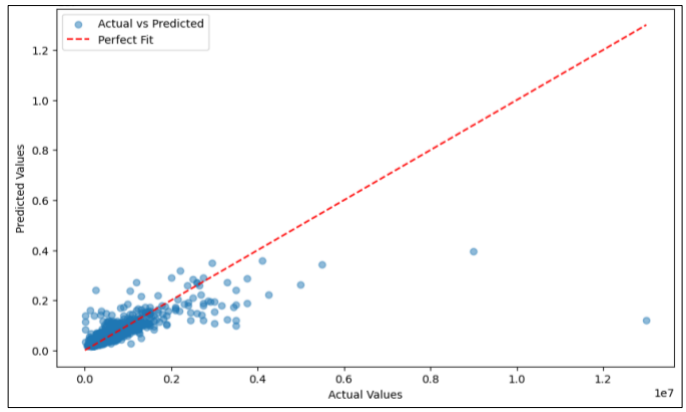


The Random Forest regression model returned a percentage of variance explained of 0.59, which shows a substantial improvement in capturing the variability in housing prices compared to the linear regression model. The RMSE value at $310,821, is also lower than the previous linear regression model, suggesting that the random forest model provides more accurate predictions on average. The residuals plot for the random forest model also indicates that this model was able to more accurately capture and better represent the dataset. Looking at the residuals plot, we see that the random forest model has less bias in its predictions, evident by the tighter clustering around y=0. This plot also shows that the random forest model has far fewer outliers than the linear regression model. Although performance is significantly improved, the presence of outliers indicates that there were still a handful of records that the model struggled to predict.

Random Forest Residuals Plot

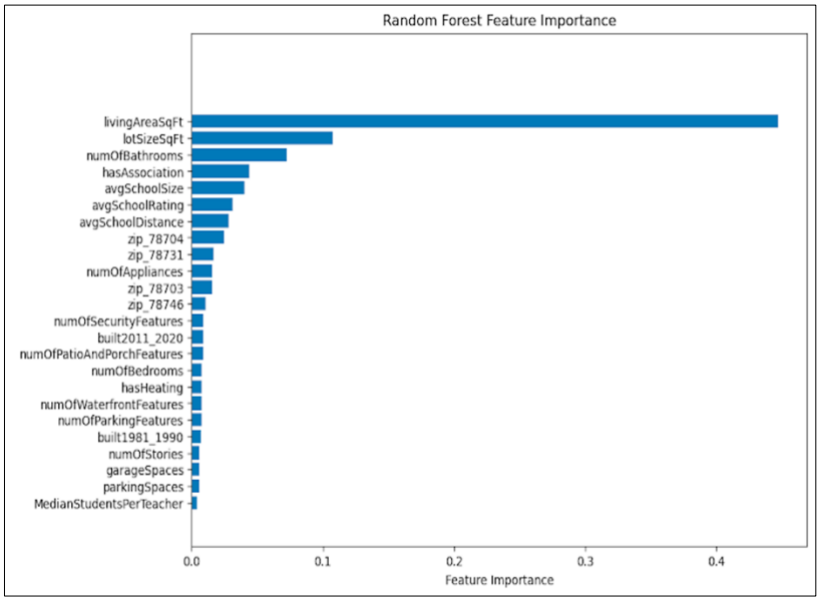


Random Forest Actual vs. Predicted Values



We then created a feature importance chart that shows us which features in the dataset gave the strongest signal in respect to the regression method. In the chart below, we can see that living area stands far above the rest. Lot size and number of bathrooms round out the top three most important features with respect to house price, closely followed by features relating to school performance, and HOA. This will be interesting to compare to the next model we test.

Random Forest - Feature Importance Breakdown



Overall, the random forest model provides better predictions and performance compared to the linear regression model but given the black box nature of the Random Forest, we decided to utilize the XGBoost model next for deeper exploration.

Our **XGBoost** approach resulted in a grid search that yielded many strong models with R-squared's between 0.59 (at number of trees=250 and max_depth=2) and 0.62 (at number of trees=250 and max_depth =4), with expected incremental reductions in Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) as the number of trees (*nrounds*) and the maximum depth of a tree (*max_depth*) grew, yet we managed to avoid overfitting as the evaluation metrics on the test set also came out equally as strong (R^2=0.72). Additionally, we got a first look at feature importances, where the Living Area (Square Footage) took the top spot by a significant magnitude. Secondarily, # of Bathrooms and Lot Size (Square Footage) came in as significant as well, followed by 3 different School system related factors - distance, school rating, and school size. Some geographic features carrying signal are the Zip Code representing South Austin (*Explore Austin's Hottest Zip Code* blog by Austin Relocation Guide) and Central Austin west of Downtown (Tarrytown, Clarksville, Old West Austin). All these influential components are in agreement with our initial intuition, though our initial models weighted # of Bedrooms more significantly. That said, in our new data set, we have other home size-related features with potentially overlapping signal with # of Bedrooms.

3-Fold XGB CV Grid Search Results

max_depth	nrounds	RMSE	Rsquared	MAE
2	250	302763.9	0.5972758	110947.7
2	500	301249.5	0.6016266	108159.9
2	750	300876.8	0.6026511	107421.2
4	250	295855.3	0.6182791	105546.5
4	500	296839.8	0.6162848	106234.4
4	750	297444.6	0.6150366	107175.2
6	250	302608.5	0.6059239	106098.5
6	500	303292.5	0.6044691	107506.7
6	750	303555.3	0.6039288	108070.3

XGB Feature Importance

livingAreaSqFt	100.0000000	numOfAppliances	3.8009857
numOfBathrooms	24.2807875	zip_78746	3.2998098
lotSizeSqFt	18.8141736	built2011_2020	2.5143374
hasAssociation	8.8848825	propertyTaxRate	2.4534355
avgSchoolDistance	8.2483403	numOfParkingFeatures	1.7460014
avgSchoolRating	8.0175088	garageSpaces	1.6510291
avgSchoolSize	6.2528284	hasView	1.4625000
zip_78704	5.6306825	hasSpa	1.4229031
zip_78703	5.4908811	numOfWaterfrontFeatures	1.3196513
zip_78731	4.2030511	numOfBedrooms	0.9255745

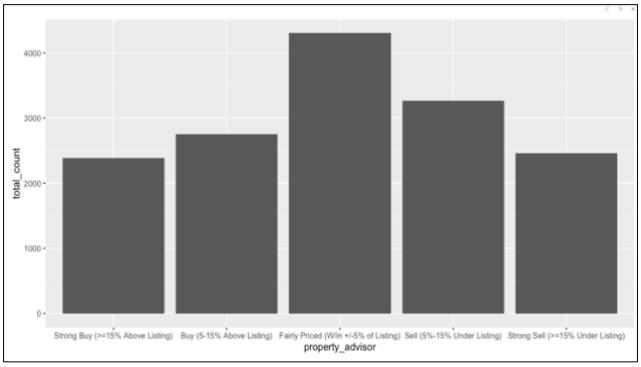
The Austin Property Deal Recommender

After analyzing the performance of our three models, we decided to create the recommendation tool using the model with the best overall performance: The XGBoost model. We envision that this tool will be productively utilized in two different real-world applications:

Property Deal Recommender for Prospective Home Buyers:

Home hunters in the Austin area can leverage our models to automate their search for a good deal in the local housing market. We decided to use our models to make price predictions on our list of homes and compare those predictions to the actual listing prices. If our model predicted a price much higher than the actual price (where the prediction is >=15% above listing price), we think of this as a good deal for a buyer and label this group *Strong Buy* (see figure below). Our intuition that Austin housing prices are quite expensive now generally holds true as we can see there are far more Sell/Strong Sell recommendations than Buys/Strong Buys. Additionally, when summarizing what the average Strong Buy looks like vs. the average Strong Sell, we can see that Strong Buys, despite being nearly half the price of Strong Sells, see higher living areas, lot sizes, bathroom counts, and are closer to higher rated schools.

Deal Recommender - Property Count by Recommendation Category



Deal Recommender - What Do Strong Buys & Strong Sells Look Like?
Top Numeric Feature Means for Strong Buy vs. Strong Sells

Top Features	**Strong Buy (Avg.)**		**Strong Sell (Avg.)**
Listing Price	**$358,284**	**<**	**$616,443**
Living Area (Square Feet)	2,158	>	2,056
# of Bathrooms	2.64	>	2.58
Lot Size (Square Feet)	27,159	>	23,570
avgSchoolDistance	1.77	>	1.72
avgSchoolRating	5.66	>	5.43
avgSchoolSize	1,194	>	1,186

Renovation Prioritizer for Homeowners Looking to Maximize the ROI of Upgrade Investments:

We can also leverage our models for existing homeowners who don't want to sell but want to increase their home equity & personal net worth. One great way to do this is to upgrade your home while ensuring that every dollar spent doing so will result in a significantly higher home price than your initial investment costs. Some locational, size, tax rates, and age features about homes are much tougher to alter, despite being significant to our pricing models. That said, some features represent opportunities for manageable upgrades, such as adding bathrooms, upgrading or adding appliances, adding parking features, garage spaces, and possibly even a spa. Some investments not as worthwhile, as their relationship with higher prices as more muted include heating/cooling features, accessibility features, patio and porch features, or security features. In a production environment, homeowners could input their properties details into our tool, alter the inputs for one or more of their potential renovation opportunities, and see how elastic the market price prediction would be. Comparing that increase to the cost of the project would be a great way to calculate a project ROI. We imagine this as a front-end web application that allows users to input their property and project cost details, access our model's prediction, resulting in a web page that yields a ROI-ranked list of renovations.

Conclusion

Throughout this project, we examined housing data from one of the nation's fastest growing metroplexes: Austin, TX, with the aim of developing a tool to help individuals make informed decisions in a complex real estate market.

The dataset that we chose, sourced from Northwestern University via Kaggle, was selected due to its large range and variety of features that provided a comprehensive summary of what homebuyers may be looking for in their search. We gravitated toward this dataset because we found that simpler datasets with fewer features had extremely strong correlations with one or two features, skewing other notable features that logically would also provide a strong signal. Before we could continue with this dataset, there was cleaning and preprocessing that needed to be done. Using our best judgment, we omitted any features that we deemed to be extraneous, such as latitude and longitude, since any meaningful signal from those features was also captured by zip code. Removing these unnecessary features allows us to enhance model accuracy and prevent overfitting. We then elected to convert any categorical variables, such as pool and HOA, into binary booleans. Next, we one hot encoded features with different factor levels, such as home type and zip code, allowing for more model flexibility. Lastly, we omitted any records with missing values.

We chose to test three different regression analysis techniques with increasing levels of complexity: multivariable linear regression, random forest regression, and extreme gradient boosting. We chose to start with a linear regression model for its simplicity, using it as a baseline for comparing models moving forward. Random forest regression, an ensemble learning method, was chosen due to its tendency to work well with

housing data, since it can capture non-linear relationships. Due to its robust nature, it is also able to better handle outliers and irregularities in the data. Lastly, the XGBoost model was chosen for its greater ability to handle imbalanced datasets and increased control over hyper parameter tuning.

In our linear regression model, we noticed that the model was only capturing about half of the variation of house prices in the dataset (R^2 = 0.47). This model also performed well regarding accuracy, having a RMSE of $353,607, which considering the magnitude of the dependent variable (house prices), is satisfactory.

The next model we created was a random forest regression model. This model was significantly better at capturing the variability in house prices (R^2 = 0.59) but was only a little bit more accurate than the linear regression model ($310,821).

Our XG Boost model saw a considerable jump in variance and prediction accuracy. With some tree depth adjustment, we returned an R^2 value of 0.62 and a RMSE of $300,876. It is important to note that we were able to avoid overfitting while tuning, proven by the congruency of the evaluation metrics when tested on the training set (R^2 = 0.72). Given the increased performance of the XG Boost model, we chose to proceed with it for our “Deal Recommender” tool.

Additionally, after performing a feature evaluation, we were able to confirm our preliminary hypotheses. The most important features were building and lot size, number of bedrooms and bathrooms, followed closely by various school factors. Armed with this information, we proceeded to build the recommendation tool.

Setting adjustable thresholds for buy/sell recommendations, we tested it with our dataset and saw that the tool suggests that there are more sell recommendations, confirming our intuition that Austin is currently a seller’s market. This tool can also be used by existing homeowners looking to increase their home worth, by focusing renovations on more important internal features, as determined by our model.

Lastly, we acknowledge that this project has its own shortcomings. The data collected ranges from 2018-2021, meaning that any housing market trends in the past two years were not captured by our model. The housing market has also been historically volatile, being influenced by uncontrollable external factors, meaning that our model would struggle to predict any sudden trends. However, with frequent and periodic updating and retraining, this tool has the potential to help the local community better understand their complex real estate market and make educated choices.

References

- U.S. Census Bureau & U.S. Department of Housing and Urban Development. (2023, November 30). Median Sales Price of Houses Sold for the United States [MSPUS]. Retrieved from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/MSPUS

- Eric Pierce. (2021). Austin Housing Prices. Kaggle. https://www.kaggle.com/datasets/ericpierce/austinhousingprices

- City of Austin. (2023, May 19). Austin Now 10th Largest City in U.S. City of Austin. https://www.austintexas.gov/news/austin-now-10th-largest-city-us

- KVUE. (2022, October 18). Austin is the 2nd fastest growing city in the U.S. KVUE. https://www.kvue.com/article/money/economy/boomtown-2040/austin-2nd-fastest-growing-city-us/269-d9bc4e7b-4e03-4380-8605-bd18b699b327

- Home Buying Institute. (2023). Austin Housing Market Forecast. Home Buying Institute. https://homebuyinginstitute.com/mortgage/austin-housing-market-forecast/